Chapter P4: Explaining motion

Overview

Simple but counterintuitive concepts of forces and motion, developed by Galileo and Newton, can transform young people's insight into everyday phenomena. These ideas also underpin an enormous range of modern applications, including spacecraft, urban mass transit systems, sports equipment and rides at theme parks.

Topic P4.1 reviews the idea of forces: identifying, describing and using forces to explain simple situations. Topic P4.2 looks at how speed is measured and represented graphically and introduces the vector quantities of velocity and displacement. The relationships between distance, speed, acceleration and time are an example of simple mathematical modelling that can be used to predict the speed and position of a moving object.

The relationship between forces and motion is developed in Topic P4.3, where resultant forces and changes in momentum are described. These ideas are then applied in the context of road safety.

Topic P4.4 considers how we can describe motion in terms of energy transfers.

Learning about forces and motion before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- describe motion using words and with distance–time graphs
- use the relationship average speed = distance ÷ time
- identify the forces when two objects in contact interact; pushing, pulling, squashing, friction, turning
- use arrows to indicate the different forces acting on objects, and predict the net force when two or more forces act on an object
- know that the forces due to gravity, magnetism and electric charge are all non-contact forces
- understand how the forces acting on an object can be used to explain its motion.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

P4.1 What are forces?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*	*Linked learning opportunities*
Force arises from an interaction between two objects, and when two objects interact, both always experience a force and that these two forces form an interaction pair. The two forces in an interaction pair are the same kind of force, equal in size and opposite in direction, and act on different objects (Newton's third law). Friction is the interaction between two surfaces that slide (or tend to slide) relative to each other: each surface experiences a force in the direction that prevents (or tends to prevent) relative movement. There is an interaction between an object and the surface it is resting on: the object pushes down on the surface, the surface pushes up on the object with an equal force, and this is called the normal contact force. In everyday situations, a downward force acts on every object, due to the gravitational attraction of the Earth. This is called its weight. It can be measured (in N) using a spring (or top-pan) balance. The weight of an object is proportional to its mass. Near the Earth's surface, the weight of a 1 kg object is roughly 10 N. The Earth's gravitational field strength is therefore 10 N/kg. Newton's insight that linked the force that causes objects to fall to Earth with the force that keeps the Moon in orbit around the Earth led to the first universal law of nature.	1. recall and apply Newton's third law 2. recall examples of ways in which objects interact: by gravity, electrostatics, magnetism and by contact (including normal contact force and friction) 3. describe how examples of gravitational, electrostatic, magnetic and contact forces involve interactions between pairs of objects which produce a force on each object 4. represent interaction forces as vectors 5. define weight 6. describe how weight is measured 7. recall and apply the relationship between the weight of an object, its mass and the gravitational field strength: weight (N) = mass (kg) × gravitational field strength (N/kg) M1c, M3b, M3c	**Practical work** • Investigate the effect of different combinations of surfaces on the frictional forces. **Ideas about science** • Explain how Newton's discovery of the universal nature of gravity is an example of the role of imagination in scientific discovery (IaS3).

P4.2 How can we describe motion?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
The motion of a moving object can be described using the speed the object is moving, the direction it is travelling and whether the speed is changing. The distance an object has travelled at a given moment is measured along the path it has taken. The displacement of an object at a given moment is its net distance from its starting point together with an indication of direction. The velocity of an object at a given moment is its speed at that moment, together with an indication of its direction. Distance and speed are scalar quantities; they give no indication of direction of motion. Displacement and velocity are vector quantities, and include information about the direction. In everyday situations, acceleration is used to mean the change in speed of an object in a given time interval. Distance–time graphs and speed-time graphs can be used to describe motion. The average speed can be calculated from the slope of a distance–time graph. The average acceleration of an object moving in a straight line can be calculated from a speed-time graph. The distance travelled can be calculated from the area under the line on a speed-time graph.	1. recall and apply the relationship: $\text{average speed (m/s)} = \frac{\text{distance (m)}}{\text{time (s)}}$ M1a, M1c, M3b, M3c, M3d 2. recall typical speeds encountered in everyday experience for wind, and sound, and for walking, running, cycling and other transportation systems 3. a) make measurements of distances and times, and calculate speeds b) describe how to use appropriate apparatus and techniques to investigate the speed of a trolley down a ramp M2b, M2f *PAG3* 4. make calculations using ratios and proportional reasoning to convert units, to include between m/s and km/h M1c, M3c 5. explain the vector–scalar distinction as it applies to displacement and distance, velocity and speed 6. a) recall and apply the relationship: $\text{acceleration (m/s}^2\text{)} = \frac{\text{change in speed (m/s)}}{\text{time taken (s)}}$ M1c, M3b, M3c, M3d b) explain how to use appropriate apparatus and techniques to investigate acceleration *PAG3*

Linked learning opportunities

Practical work:
- Use a variety of methods to measure distances, speeds and times and to calculate acceleration.
- Compare methods of measuring the acceleration due to gravity.

Ideas about Science
- Use mathematical and computational models to make predictions about the motion of moving objects (IaS3).
- Explore using simple computer models to predict motion of a moving object.

P4.2 How can we describe motion?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*	*Linked learning opportunities*
The mathematical relationships between acceleration, speed, distance, and time are a simple example of a computational model. The model can be used to predict the speed and position of an object moving at constant speed or with constant acceleration.	7. select and apply the relationship: (final speed (m/s))^2 – (initial speed(m/s))^2 = 2 × acceleration (m/s^2) × distance (m) M1a, M1c, M3b, M3c, M3d	
	8. draw and use graphs of distances and speeds against time to determine the speeds and accelerations involved	
	9. interpret distance–time and velocity–time graphs, including relating the lines and slopes in such graphs to the motion represented M4a, M4b, M4c, M4d	
	10. interpret enclosed areas in velocity – time graphs M4a, M4b, M4c, M4d, M4f	
	11. recall the value of acceleration in free fall and calculate the magnitudes of everyday accelerations using suitable estimates of speeds and times	

P4.3 What is the connection between forces and motion?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
When forces act on an object the resultant force is the sum of all the individual forces acting on it, taking their directions into account. **If a resultant force acts on an object, it causes a change in momentum in the direction of the force.** **The size of the change in momentum of an object is proportional to the size of the resultant force acting on the object and to the time for which it acts (Newton's second law).** For an object moving in a straight line: a. if the resultant force is zero, the object will move at constant speed in a straight line (Newton's first law). b. if the resultant force is in the direction of the motion, the object will speed up (accelerate). c. if the resultant force is in the opposite direction to the motion, the object will slow down. **In situations involving a change in momentum (such as a collision), the longer the duration of the impact, the smaller the average force for a given change in momentum.** In situations where the resultant force on a moving object is not in the line of motion, the force will cause a change in direction. **If the force is perpendicular to the direction of motion the object will move in a circle at a constant speed – the speed doesn't change but the velocity does. For example, a planet in orbit around the Sun – gravity acts along the radius of the orbit, at right angles to the planet's path.**	1. describe examples of the forces acting on an isolated solid object or system 2. describe, using free body diagrams, examples where several forces lead to a resultant force on an object and the special case of balanced forces (equilibrium) when the resultant force is zero *qualitative only* **3. use scale drawings of vector diagrams to illustrate the addition of two or more forces, in situations when there is a net force, or equilibrium** ⓘ ***Limited to parallel and perpendicular vectors only*** M4a, M5a, M5b **4. recall and apply the equation for momentum and describe examples of the conservation of momentum in collisions:** **momentum (kg m/s) = mass (kg) × velocity (m/s)** M1c, M3b, M3c, M3d **5. select and apply Newton's second law in calculations relating force, change in momentum and time:** **change in momentum (kg m/s) = resultant force (N) × time for which it acts (s)** M1c, M3b, M3c, M3d 6. apply Newton's first law to explain the motion of objects moving with uniform velocity and also the motion of objects where the speed and/or direction changes **7. explain with examples that motion in a circular orbit involves constant speed but changing velocity** ***qualitative only***

Linked learning opportunities
Practical work • Investigate factors that might affect human reaction times. • Investigate the use of crumple zones to reduce the stopping forces.

P4.3 What is the connection between forces and motion?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*	*Linked learning opportunities*
In some situations a resultant force acts to make an object rotate about a fixed point (pivot). The rotational effect is called the moment of the force; the further the force acts from the pivot, the greater the turning effect. Levers and gears are used to transmit rotational forces.	8. describe examples in which forces cause rotation *(separate science only)* 9. define and calculate the moment of examples of rotational forces using the equation: moment of a force (N m) = force (N) × distance (m) (normal to direction of the force) *(separate science only)* M1c, M3b, M3c, M3d 10. explain, with examples, how levers and gears transmit the rotational effects of forces *(separate science only)*	**Practical work** • **Investigate forces that cause rotation, including the use of levers and gears.**
The mass of an object can be thought of as the amount of matter in an object – the sum of all the atoms that make it up. Mass is measured in kilograms. The mass of an object is also a measure of its resistance to any change in its motion (its inertia); using this definition the inertial mass is the ratio of the force applied to the resulting acceleration. **Newton wrote about how the length of time a force acted on an object would change the object's 'amount of motion', and the way he used the term makes it clear that he is describing what we now call momentum, this has led to Newton's second law being expressed in two ways – in terms of change in momentum and in terms of acceleration.** Newton's explanation of motion is one of the great intellectual leaps of humanity. It is a good example of the need for creativity and imagination to develop a scientific explanation of something that had been observed and discussed for many years (IaS3).	**11. explain that inertial mass is a measure of how difficult it is to change the velocity of an object and that it is defined as the ratio of force over acceleration** 12. recall and apply Newton's second law relating force, mass and acceleration: force (N) = mass (kg) × acceleration (m/s^2) M1c, M3b, M3c, M3d 13. Use and apply equations relating force, mass, velocity, acceleration, and **momentum** to explain relationships between the quantities M3b, M3c, M3d	**Ideas about Science** • Explain why Newton's explanation of motion is an example of the need for creative thinking in developing new scientific explanations (IaS3).

P4.3 What is the connection between forces and motion?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
Ideas about force and **momentum** can be used to explain road safety measures, such as stopping distances, car seatbelts, crumple zones, air bags, and cycle and motorcycle helmets. Improvements in technology based on Newton's laws of motion (together with the development of new materials) have made all forms of travel much safer.	14. explain methods of measuring human reaction times and recall typical results
	15. explain the factors which affect the distance required for road transport vehicles to come to rest in emergencies and the implications for safety M2c
	16. explain the dangers caused by large decelerations **and estimate the forces involved in typical situations on a public road**
	17. given suitable data, estimate the distance required for road vehicles to stop in an emergency, and describe how the distance varies over a range of typical speeds *(separate science only)* M1c, M1d, M2c, M2h, M3b, M3c
	18. in the context of everyday road transport, use estimates of speeds, times and masses to calculate the accelerations and forces involved in events where large accelerations occur *(separate science only)* M1d, M2b, M2h, M3c

Linked learning opportunities

Ideas about Science

- Describe and explain examples of how application of Newton's laws of motion have led developments in road safety (IaS4).
- Discuss people's willingness to accept risk in the context of car safety and explain ways in which the risks can be reduced (IaS4).

P4.4 How can we describe motion in terms of energy transfers?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
Energy is always conserved in any event or process. Energy calculations can be used to find out if something is possible and what will happen, but not explain why it happens. The store of energy of a moving object is called its kinetic energy. As an object is raised, its store of gravitational potential energy increases, and as it falls, its gravitational potential energy decreases. When a force moves an object, it does work on the object, energy is transferred to the object; when work is done by an object, energy is transferred from the object to something else, for example: • when an object is lifted to a higher position above the ground, work is done by the lifting force; this increases the store of gravitational potential energy. • when a force acting on an object makes its velocity increase, the force does work on the object and this results in an increase in its store of kinetic energy. If friction and air resistance can be ignored, an object's store of kinetic energy changes by an amount equal to the work done on it by an applied force; in practice air resistance or friction will cause the gain in kinetic energy to be less than the work done on it by an applied force in the direction of motion, because some energy is dissipated through heating.	1. describe the energy transfers involved when a system is changed by work done by forces including: a) to raise an object above ground level b) to move an object along the line of action of the force
	2. recall and apply the relationship to calculate the work done (energy transferred) by a force: work done (N m or J) = force (N) × distance (m) (along the line of action of the force) M1a, M3b, M3c, M3d
	3. recall the equation and calculate the amount of energy associated with a moving object: kinetic energy (J) = $\frac{1}{2}$ × mass (kg) × (speed (m/s))^2 M1a, M3b, M3c, M3d
	4. recall the equation and calculate the amount of energy associated with an object raised above ground level: gravitational potential energy (J) = mass (kg) × gravitational field strength (N/kg) × height (m) M1a, M3b, M3c, M3d
	5. make calculations of the energy transfers associated with changes in a system, recalling relevant equations for mechanical processes M1a, M1c, M3c
	6. calculate relevant values of stored energy and energy transfers; convert between newton metres and joules M1c, M3c

Linked learning opportunities

Specification links:
- **P2 Sustainable energy**

Practical work
- Use datalogging software to calculate the efficiency of energy transfers when work is done on a moving object.
- Measure the work done by an electric motor lifting a load, and calculate the efficiency.

P4.4 How can we describe motion in terms of energy transfers?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
Calculating the work done when climbing stairs or lifting a load, and the power output, makes a link back to the usefulness of electrical appliances for doing many everyday tasks.	7. describe all the changes involved in the way energy is stored when a system changes, for common situations: including an object projected upwards or up a slope, a moving object hitting an obstacle, an object being accelerated by a constant force, a vehicle slowing down
	8. explain, with reference to examples, the definition of power as the rate at which energy is transferred (work done) in a system
	9. recall and apply the relationship: $\text{power (W)} = \dfrac{\text{energy transferred (J)}}{\text{time (s)}}$ M1a, M3b, M3c, M3d

Linked learning opportunities